S1.4 Counting particles by mass: The mole

Molar concentration of a solution

Learning outcomes

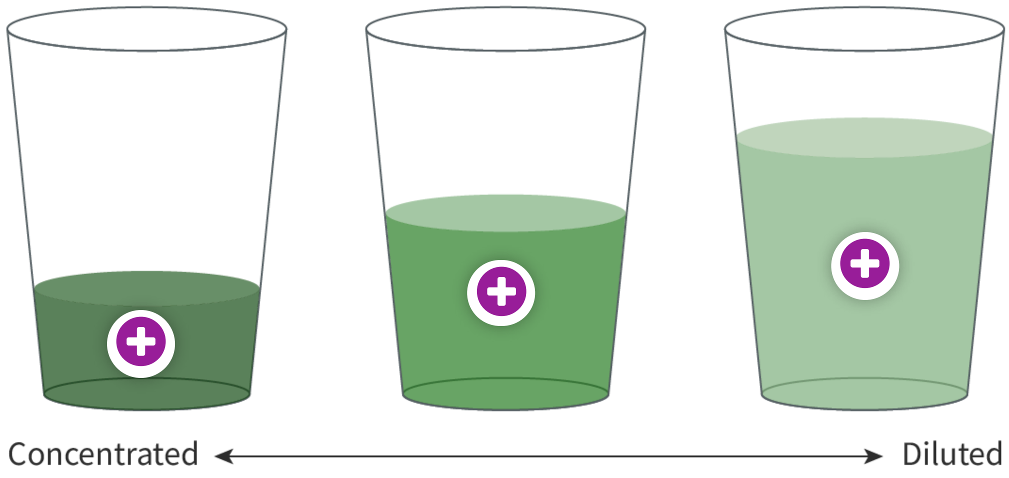
By the end of this section you should be able to:

- Describe solutions qualitatively as dilute and concentrated.
- Solve problems involving molar concentration, amount of solute and volume of solution.
- Identify laboratory equipment required to accurately prepare a solution of known concentration.

Recall from section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) that matter can be classified as a pure substance or a mixture. So far, you have learned how to calculate the number of elementary entities for pure substances, but what about mixtures? The most common mixture you will encounter in chemistry is a homogenous mixture made by dissolving a solute into a solvent, forming an aqueous solution. You have encountered aqueous solutions before in everyday life. Adding salt or sugar to water and stirring until you can no longer see the crystals is an example of an aqueous solution. Even the water you drink is not made up of only water molecules – there are many dissolved ions and even oxygen molecules dissolved in the water – you cannot see them, but they are there!

In a mixture, we express the quantity of the solute dissolved in the solution as a ratio, known as the concentration. You might be familiar with qualitative descriptions for describing the concentration of a solution. For example, if you have ever prepared a drink from a syrup or have made lemonade from lemon juice and sugar, you have taken a concentrated form of the drink that has a very strong flavour and added water to make a diluted form of the drink that has a weaker flavour. Concentrated and dilute are qualitative terms to describe concentration, but we require quantitative measurements so that we can relate concentration to the number of elementary entities. Can you think of some reasons why the precise concentration of a solution might be important?

Try **Interactive 1** to see how the concentration can be changed by the addition of solute and solvent.



Interactive 1. Dilute or Concentrated?

More information for interactive 1

An interactive illustration features three glasses containing solutions of different concentrations. The first glass is quarter-filled, the second glass is half-filled while the third glass is filled up to three-fourths with the solution. The first glass holds the most concentrated solution, indicated by its dark green color. The second glass contains a less concentrated solution than the first, indicated by its medium green tone. The third glass has a more diluted solution, indicated by its light green color.
There is a double-headed arrow at the bottom between the terms concentration and dilution, with concentration labeled on the left, pointing toward the first glass, and dilution labeled on the right, pointing toward the third glass.
Each glass has a plus icon, representing interactive hotspots. Users can click on these hotspots to reveal close-up views of the corresponding solutions. Clicking on the hotspot shows the close-up view of the corresponding glass, with the semantic zooming of the solute and solvent particles in that solution. The solvent particles are represented by small white circles while the solute particles are represented by small green circles. There is also text displayed beside the solvent and solute particles for each hotspot.
The text displayed for each hotspot is as follows:
Hotspot on first glass: Solvent particles, Solute Particles (17)
Hotspot on second glass: Solvent particles, Solute particles (16)
Hotspot on third glass: Solvent particles, Solute particles (15)
The number in parentheses beside the solute particles represents the number of solute

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particles in that solution.
This interactive helps users understand the concept of solution concentration and how it varies with the amount of solute present.

Vinegar is a product that you might have in your house because it is a very versatile substance. It can be used in cooking to add flavour to sauces and salads, as a cleaning product for glass and ceramic tiles and can even be used to relieve the sting of an insect bite. Vinegar is an aqueous solution containing ethanoic acid, also known as acetic acid, a substance which you likely also have in your school laboratory. Laboratory grade ethanoic acid is very concentrated and must be handled with extreme caution, but household vinegar is diluted to the point that it is safe to ingest or make contact with the skin. The difference is the concentration **(Figure 1)**.

Credit: Kittisak Kaewchalun, Getty Images

(https://www.gettyimages.co.uk/detail/photo/acetic-acid-in-bottle-chemical-in-the-laboratory-royalty-free-image/1474288946)

Credit: Eskay Lim / EyeEm, Getty Images

(https://www.gettyimages.co.uk/detail/photo/close-up-of-variousvinegars-on-table-against-white-royalty-free-image/963037322)

Figure 1. Household vinegar is a diluted solution of acetic acid.

Examine the labels on a few different items in your house – medication, cleaning products and toiletries. Notice whether the quantities of the active ingredients are listed on the label. When concentration is listed on a label, it must be expressed using measurements that most consumers would understand, such as grams and litres, and is often expressed as a percentage concentration $\left(\%\frac{\text{volume}}{\text{volume}} \text{ or } \%\frac{\text{weight}}{\text{volume}}\right)$. Recall that one litre is also expressed as a cubic decimetre, dm^3.

Solutions are prepared by dissolving a solute in a solvent. We accurately measure the mass of the solute and the total volume of the solution, one way of expressing solution concentration is mass per volume. For example, if a salt solution contains 1.00 g of salt dissolved in 1.00 dm^3 of solution, the concentration would be expressed as 1.00 g dm^{-3} (**Figure 2**).

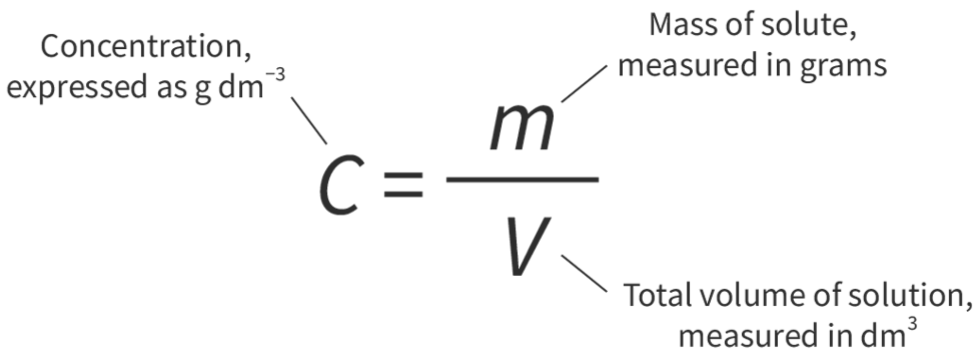

Figure 2. Concentration is expressed as mass of solute per unit volume of solution.

More information for figure 2

The image displays a formula for concentration represented as C = m/v, where 'C' denotes concentration, 'm' represents the mass of the solute, and 'v' indicates the volume of the solution. Around the formula, there are annotations explaining each component: "Concentration, expressed as g dm^—3," pointing to 'C'; "Mass of solute, measured in grams," pointing to 'm'; and "Total volume of solution, measured in dm^3," pointing to 'v'. These annotations clarify that the concentration is expressed in grams per cubic decimeter, the mass is in grams, and the volume is in cubic decimeters.

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Try using the concentration formula yourself in these worked examples.

Worked example 1

Calculate the mass of sodium chloride, NaCl, required to make 2.00 dm^3 of 0.800 g dm^{-3} solution.

Solution steps	Calculatior
Step 1: first, identify the formula to use.	$C = \frac{m}{V}$
Step 2: rearrange the formula to make *m* the subject.	$m = C \times V$
Step 3: substitute in the known values.	$m = 0.800$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	$= 1.60$ g N
Step 5: state the answer with appropriate units and the number of significant figures used in rounding	Therefore, prepare thi

Worked example 2

Calculate the volume of solution that can be prepared when 300.0 mg of citric acid is diluted to a concentration of 0.500 g dm^{-3} solution.

Solution steps	Calcula
Step 1: first, identify the formula to use.	$C = \frac{m}{V}$

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Solution steps	Calcula
Step 2: rearrange the formula to make V the subject.	$V = \frac{m}{C}$
Step 3: convert solute from mg to g.	$\frac{300}{1000} =$
Step 4: substitute in the known values.	$V = \frac{0.}{0.}$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding.	$= 0.600$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	Therefo can be

As a chemist, you must also be able to express concentration in terms of moles per unit volume, since we use moles as a way of grouping and counting elementary entities. Recall in section S1.4.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/) that you learned how to calculate the amount in moles from a mass in grams. **Figure 3** has the formula for calculating the concentration of a solution in mol dm^{-3}.

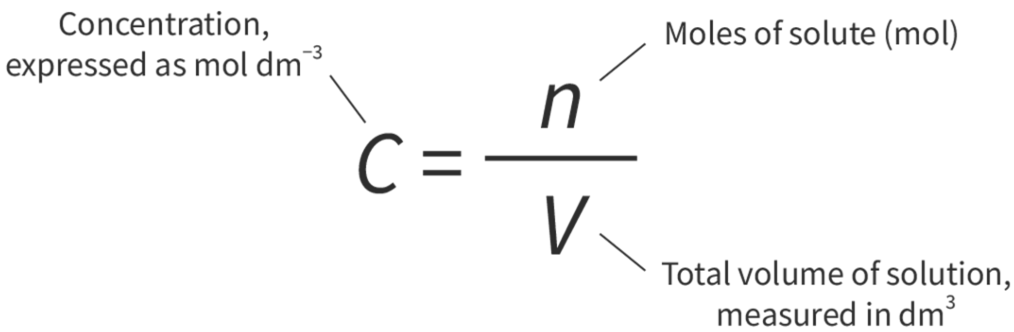

Figure 3. Concentration can also be expressed in moles per unit volume.

More information for figure 3

The image is a diagram showing the formula for calculating concentration in terms of moles per unit volume. It displays the mathematical expression: C = n / V. Three annotations are present:

1. 'C' is labeled as Concentration, expressed as mol dm^-3.
2. 'n' denotes the Moles of solute (mol).
3. 'V' represents the Total volume of solution, measured in dm^3.

This formula indicates that concentration (C) is calculated by dividing the number of moles of solute (n) by the volume of the solution (V) in cubic decimeters (dm^3).

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Worked example 3

Calculate the molar concentration, in mol dm^{-3}, of the 0.500 g dm^{-3} citric acid, $C_6H_8O_7$ solution.

Solution steps	Calculations
Step 1: first, calculate the molar mass of citric acid.	$M = (6\times12.01) + (8\times1.($ $= 192.14\ \text{g mol}^{-1}$
Step 2: calculate the amount (in mol) of citric acid	$\frac{0.500}{192.14} = 0.00260\ \text{mol}$
Step 3: divide the amount (in mol) by the volume (in dm^3)	$\frac{0.00260}{1} = 0.00260\ \text{mol}$ (
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	The molar concentration c $0.00260\ \text{mol dm}^{-3}$.

The use of square brackets, [ ], is often used to express molar concentration, also known as molarity. The formula of the solute is placed in the square brackets. For example, a 1.00 mol dm^{-3} solution of sodium chloride, NaCl, can be expressed as:

$$[NaCl] = 1.00\ \text{mol dm}^{-3}$$

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Worked example 4

Calculate the molar concentration of a solution that is prepared by dissolving 5.00 g of sodium chloride, NaCl, in 2.00 dm^3 of water.

Solution steps	Calculations
Step 1: first, convert the mass of sodium chloride into amount (in mol) using the molar mass.	$n = \frac{5.00}{58.45}$ $= 0.0855 \text{ mol NaCl}$
Step 2: identify the formula to be used.	$n = CV$
Step 3: rearrange to make C the subject.	$C = \frac{n}{V}$
Step 4: substitute the known values.	$C = \frac{0.0855}{2.00}$ $= 0.0428 \text{ mol dm}^{-3}$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding.	Therefore, the concent $0.0428 \text{ mol dm}^{-3}$, or it $[\text{NaCl}] = 0.0428 \text{ mol d}$

Study skills

The formula *n* = *CV* is included in the DP chemistry data booklet, so you do not have to memorise it. When concentration is expressed using moles, it is referred to as molar concentration or molarity. The units are expressed as mol dm^{-3}, mol L^{-1} or simply M, where M represents molar concentration, or molarity. For example, a 1.00 M solution is said to have a concentration of 'one molar'. Your exam paper questions will always show units of mol dm^{-3}, but you may show your work or display an answer containing units of mol/L or M on your exam paper or on your laboratory work.

Worked example 5

Determine the mass of magnesium bromide, $MgBr_2$, required to make 200.0 cm^3 of 0.0500 mol dm^{-3} solution.

This calculation requires two steps – first to determine the amount of the solute, in mol, required for the solution, then converting that to an amount in grams.

Solution steps	Calculations
Step 1: identify the formula to be used.	$n = CV$
Step 2: change the units of volume from cm^3 to dm^3.	$\frac{200}{1000} = 0.200\ \mathrm{dm}^3$
Step 3: substitute the known values.	$n = 0.0500 \times 0.2000$ $= 0.0100\ \mathrm{mol\ MgBr_2}$ The amount of $\mathrm{MgBr_2}$ ir solution is $0.0100\ \mathrm{mol}$, s in grams.
Step 5: calculate the molar mass.	$M\,(\mathrm{MgBr_2}) = (1 \times 24.$ $= 184.11$
Step 6: calculate the mass of the compound.	$\mathrm{mass} = 0.0100 \times 184.1$ $= 1.84\ \mathrm{g}$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	The mass of $\mathrm{MgBr_2}$ re solution is $1.84\ \mathrm{g}$.

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Practical skills

Tool 1: Experimental techniques — Measuring variables

Preparing a standard solution of a specific concentration is a skill that requires careful measuring and special equipment.

Preparing a standard solution

When you are sick and require medication, you want to be certain that you are taking the correct dose – if the quantity is too low, it will not be effective in treating your illness and if it is too high, the medication can cause toxic effects in the body. You might remember as a small child that oral medication is often given in a liquid form, rather than in pill form, because it is easier for children to swallow. The doctor calculates the dosage required and the pharmacist prepares a suspension of the active medicinal ingredient in a syrup. If either the doctor or the pharmacist makes an error, the medicine will not be effective in treating the illness. Likewise, preparing solutions for chemical reactions requires careful measuring to be sure the concentration is correct.

The mass of the solute must be measured accurately on a mass balance – to the nearest 0.01 or 0.001 g. It is then combined with a small amount of distilled water before being transferred to a volumetric flask. A volumetric flask is a specialised piece of glassware that measures precisely to one volume measurement only (**Figure 4**).

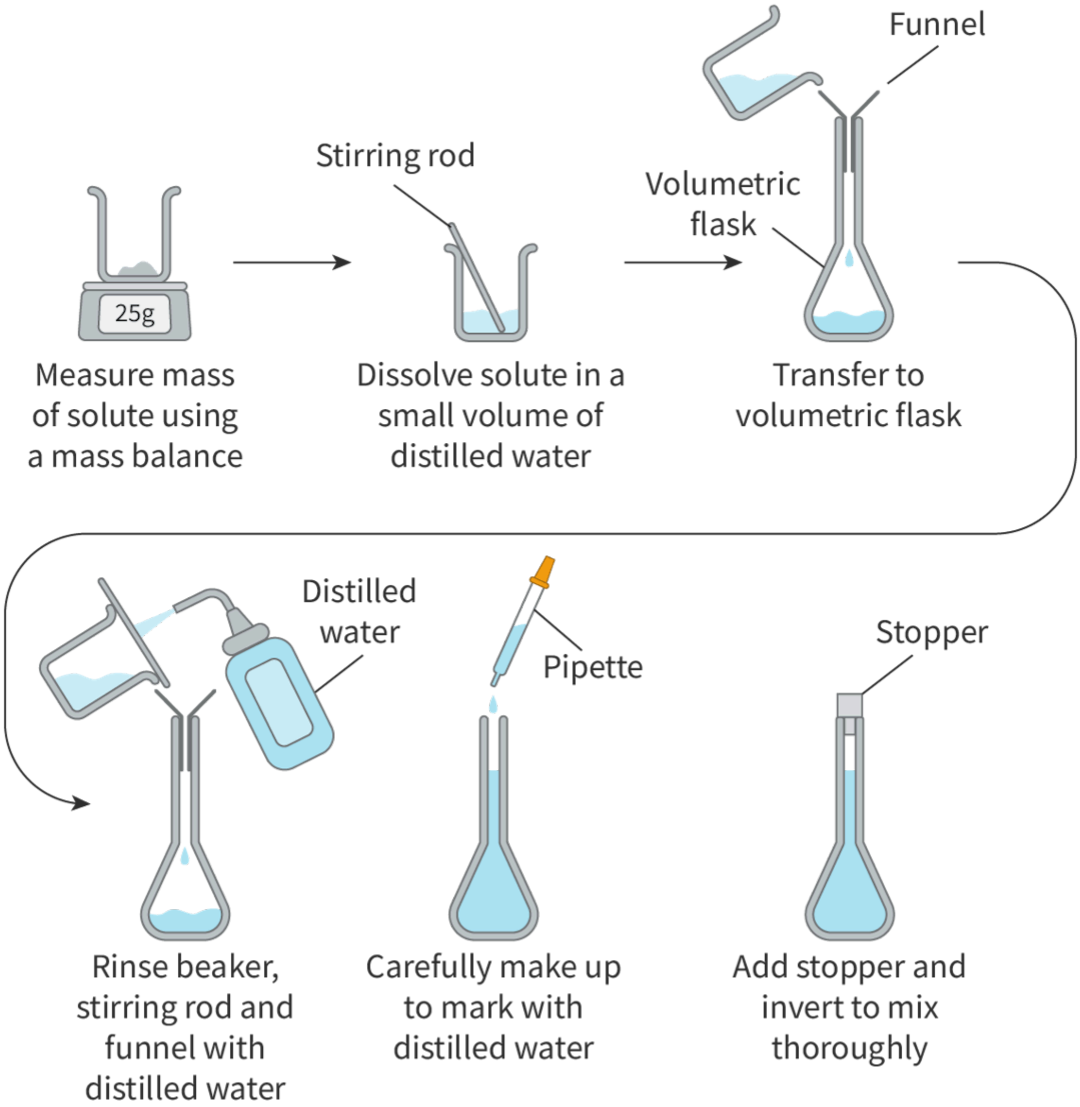

Figure 4. Preparation of a standard solution requires careful measuring to ensure an accurate concentration.

More information for figure 4

The flowchart illustrates the process of preparing a standard solution in several steps. It begins with measuring the mass of the solute using a mass balance, indicating a value of 25g. Next, the solute is dissolved in a small volume of distilled water using a stirring rod. This solution is then transferred to a volumetric flask through a funnel.

After transferring, the procedure involves rinsing the beaker, stirring rod, and funnel with distilled water to ensure all solute is transferred. Then, using a pipette, distilled water is carefully added to make up to the mark on the volumetric flask. Finally, a stopper is added, and the flask is inverted to mix the solution thoroughly, completing the preparation process.

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Practical skills

Tool 1: Experimental techniques — Measuring variables

Another method of preparing a solution involves making a dilution using a solution that is already prepared. This is sometimes referred to as a making a 'secondary' solution using a more concentrated 'primary' solution. Since the dilution is made by adding more solvent and the amount of solute does not change, the amount in mol remains the same for both the primary and secondary solutions.

Using $n = C_1V_1$ for the primary solution and $n = C_2V_2$ for the secondary solution, we can combine these two equations into one, setting the number of moles equal.

$$C_1V_1 = n = C_2V_2 \text{ , therefore } C_1V_1 = C_2V_2$$

For example, 1.00 dm^3 of a $0.300 \text{ mol dm}^{-3}$ solution of NaOH was diluted to a volume of 2.00 dm^3. Determine the concentration of the diluted solution.

$$\begin{aligned} C_1V_1 &= C_2V_2 \\ 0.300 \times 1.00 &= C_2 \times 2.00 \\ C_2 &= 0.150 \text{ mol dm}^{-3} \end{aligned}$$

We can see that the concentration of the secondary solution is less than the primary solution, which makes sense because it has been diluted.

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

Access to clean water is number 6 (https://www.un.org/sustainabledevelopment/water-and-sanitation/) of the United Nations' Sustainable Development Goals (SDGs). According to the UN, more than 800,000 people die every year from a lack of access to clean drinking water and inadequate sanitation. Reasons for this include climate change, water pollution and mismanagement of available water supplies. Contaminants (https://www.epa.gov/ccl/types-drinking-water-contaminants)

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found in polluted water can include physical contaminants such as rubbish (trash), chemical contaminants such as pesticides and biological contaminants such as harmful microorganisms.

As an IB student and a global citizen, what can you do to help those people affected by a lack of access to clean drinking water? One possible CAS experience could be to raise awareness of the situation through a poster campaign at your school or in your local community. What other ways can you think of to help achieve these goals?

Use the activity in the next section to investigate molarity.